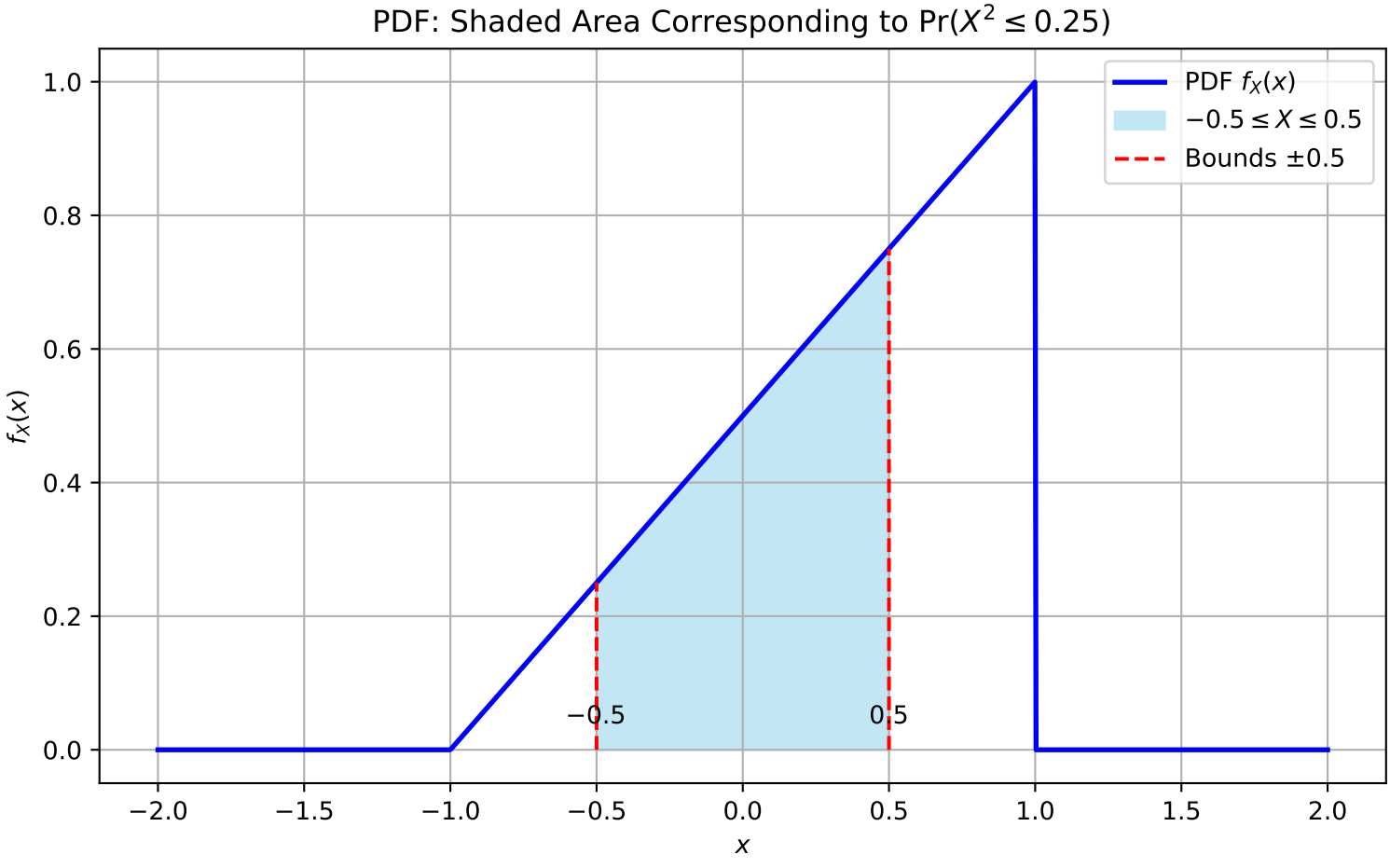
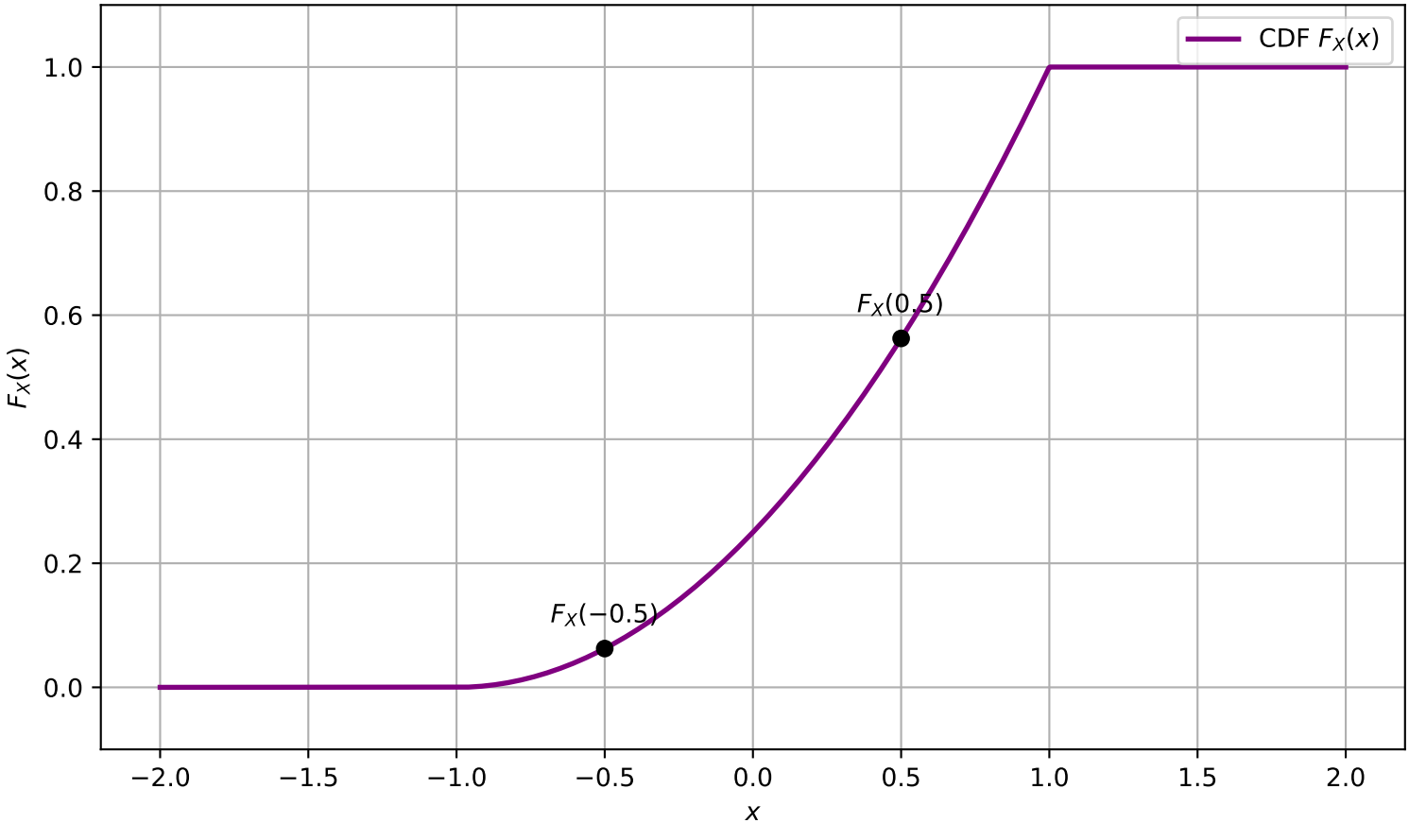

PDF: Shaded Area Corresponding to Pr($X^2 \leq 0.25$)
PDF $f_X(x)$
$-0.5 \leq X \leq 0.5$
Bounds ± 0.5
-0.5
0.5
$f_X(x)$
x


CDF: Pr($X^2 \leq 0.25$) = $F_X(0.5) - F_X(-0.5) = 0.5$
CDF $F_X(x)$
$F_X(0.5)$
$F_X(-0.5)$
$F_X(x)$
x